

3. Comparison of Design Patterns

Criteria	Composite Pattern	Flyweight Pattern
Purpose	Organizes tree structures	Reduces memory usage
Best For	Rooms and devices hierarchy	Repeated shared objects
Main Focus	Easy group control	Memory optimization
Suitability	Very suitable	Less suitable

Although the Flyweight Pattern is useful for reducing memory consumption in systems with many repeated objects, it does not directly address the hierarchical control problem required in this Smart Home System.